

4. Infinitesimal automorphisms: proof of 2.6(i)

A stable supercurve has no non-trivial infinitesimal automorphism.

Locally, on the smooth locus, the sheaf $\mathcal{A}$ of infinitesimal automorphisms projects isomorphically onto $T_X/\mathcal{D}$, where $\mathcal{D}$ denotes the structural distribution. For

$$X = (C, \mathcal{O}_C \oplus \omega_C^{1/2}), \quad \mathcal{A} = \omega_C^{-1} \oplus \omega_C^{1/2} \otimes \omega_C^{-1}.$$

The first task, near a singularity, is to compute which meromorphic sections of $\mathcal{A}$ are infinitesimal automorphisms. One has to send Θ into Θ .

Case I. Here Θ_X^- is a square root of ω_C . Let D be as in 3.1, I, where it was denoted D_0 . An infinitesimal automorphism is written as

$$aD + bD^2 \quad \text{with} \quad (-1)^{\deg a} 2a + Db = 0,$$

or equivalently

$$a = \mp \frac{1}{2} (-1)^{\deg b} Db.$$

In order for it to send Θ into Θ , one needs:

$$\begin{aligned} \text{even derivation : } & b \text{ even, } b \in \mathcal{O}^+, \\ \text{odd derivation : } & b \text{ odd, } b \in \mathcal{O}^-. \end{aligned}$$

Thus, in a neighbourhood of a component C ,

$$\boxed{\mathcal{A} = \omega_C^{-1} \oplus \omega_C^{1/2} \otimes \omega_C^{-1}}.$$

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Case II. Similarly, take D as in 3.1, II, and write the derivations as

$$\begin{aligned} \text{even : } & \frac{1}{2} Db \cdot D + bD^2 \quad (b \text{ even}), \\ \text{odd : } & -\frac{1}{2} Db \cdot D + bD^2 \quad (b \text{ odd}). \end{aligned}$$

The condition for b , a rational section of $\mathcal{O}$, to give rise to a derivation $\Theta \rightarrow \Theta$, is

$$\begin{aligned} \text{even : } & b \in \mathcal{O}^+, \quad b \text{ vanishes at } 0, \\ \text{odd : } & b \in \mathcal{O}^-, \quad b \text{ vanishes at } 0. \end{aligned}$$

In terms of the branches B_i , one has

$$\mathcal{A} = \mathcal{A}_1 \oplus \mathcal{A}_2, \quad \mathcal{A}_i = T_{B_i}(-0) \oplus T_{B_i}(-0) \otimes \omega_{B_i}^{1/2}.$$

¹On the smooth locus, $[D, X + aD] \bmod \mathcal{D} = DX + (-1)^{|a|} a \partial_z$; thus there is a unique a that brings the bracket back into $\mathcal{D}$.

An automorphism $\sim \text{id}$, even or odd, therefore gives, on the normalization of a component A of C , a non-zero section of a line bundle of degree

$$\begin{aligned} \text{even : } & (2 - 2g) - \frac{1}{2}(\# \text{ points of type I}) - (\# \text{ points of type II}), \\ \text{odd : } & (1 - 2g) - \frac{1}{2}(\# \text{ points of type I}) - (\# \text{ points of type II}). \end{aligned}$$

Under the stability hypothesis, these degrees are strictly negative. Hence there is no non-trivial infinitesimal automorphism.

5. Valuative criterion of properness

Here the only parameter spaces one has to consider are traits (S, η, s) , with $S = \text{Spec}(V)$, and one can therefore use the point of view of 1.5. By Deligne–Mumford, the ordinary stable curve C_η degenerates uniquely into a stable curve C over S , after ramifying S . What remains, after a further ramification if necessary, is to make L_η and the pairing with values in $\omega_{C/S}$ degenerate.

Away from the singular points, one can extend L_η to an invertible sheaf L . Correcting L by a divisor concentrated on the special fibre, and after passing to a double cover of S , one may assume that

$$L^{\otimes 2} \longrightarrow \omega_{C/S}$$

is an isomorphism away from the singular points. After ramification, the unique extension of this L will do.

In an étale neighbourhood of each singular point, the square roots of ω_X on the complement of the singular point are classified by

$$H^1(\mu_2) = H^1(\mathbf{Z}/2).$$

Choose a section s . In L , over $X - s$, the $\sqrt{z}$'s form a double cover. The variety of vanishing cycles has an $H^1(\mathbf{Z}/2) \simeq \mathbf{Z}/2$. After ramification over the base S and étale localization at the singular point, there are therefore two possible cases. They are covered by cases I and II of 2.3.

Yours,

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